

Big Project

December 1, 2025

Problem description

A company with several buildings across the country operates, at each site, a drone that analyzes the exterior surfaces of the building using cameras and various types of sensors. For each building, the set of points at which the drone must stop to perform measurements is known. Each point in the given set is identified by its three spatial coordinates; we may neglect the issue of drone orientation at each point.

The drone departs from an initial point (x_0, y_0, z_0) , which must also be determined (it can be chosen from a set specified below) and where the base with the charging station will be installed. It then explores a subset of the points, respecting its remaining battery charge, returns to the starting point, and—after recharging—makes further trips to visit the remaining points, so as to cover all points in one or more trips.

The speeds of the drone are:

- 1 m/s when ascending;
- 2 m/s when descending;
- 1.5 m/s for horizontal movements;
- a combination for oblique movements: for a segment with lateral movement of length a and vertical (upward) movement of length b , the travel time is

$$\max \left\{ \frac{a}{1.5 \text{ m/s}}, \frac{b}{1 \text{ m/s}} \right\}.$$

We assume that the drone is either stationary or moving at the speeds listed above; in other words, we may neglect the acceleration and deceleration phases required to move from one point to another.

The drone has a battery whose capacity depends on the building and whose charge is consumed as follows (the consumption per meter follows from the velocities listed above, according to direction):

- 50 Joule (J) per meter when ascending;
- 5 J/m when descending;
- 10 J/m for horizontal movements;
- for movements involving both a change in the z coordinate and horizontal displacement, the energy consumption is the sum of the relevant components.

We must determine the starting point and the drone trajectories for all its trips so that:

1. every point of the given grid is *covered*, i.e., the drone must pass over each grid point (including the entry points) at least once;
2. the drone returns to the starting point for recharging whenever necessary;
3. the total flight time is minimized.

Two points A and B of the grid are *connected* (i.e., the drone can travel from A to B and vice versa) if and only if:

- the Euclidean distance between A and B is at most 4 m, *or*
- the Euclidean distance between A and B is at most 11 m and *two* among the coordinates x , y , and z differ by at most 0.5 m.

This condition does *not* apply to the segments between the base point and the “entry” points of the grid, specified below, which are the only points accessible to the drone when departing from the base and when returning to it. The drone may travel along the segments between the base and any of these entry points regardless of distance.

Two instances of the problem are provided in the attached files `Edificio1.csv` and `Edificio2.csv`, in CSV format, containing the spatial coordinates (x, y, z) of the points to be visited. The sets of coordinates for the starting point (x_0, y_0, z_0) are specified below.

- For `Edificio1.csv`: the base point (x, y, z) must be chosen among integer-coordinate points with $-8 \leq x \leq 5$, $-17 \leq y \leq -15$, and $z = 0$; the entry points are all grid points (x, y, z) with $y \leq -12.5$; the drone battery capacity is 1 Wh.
- For `Edificio2.csv`: the base point (x, y, z) must be chosen among integer-coordinate points with $-10 \leq x \leq 10$, $-31 \leq y \leq -30$, and $z = 0$; the entry points are all grid points (x, y, z) with $y \leq -20$; the battery capacity is 6 Wh.

Once a solution has been obtained within a time limit of two hours, it must be visualized using a Python library such as `matplotlib`. Each trip should be plotted in a different color.

Operational instructions for the submission

Upload a file named `Cognome_Nome_IDpersona.zip`, using your surname, name, and personal ID (not your student number). This must be a `.zip` archive containing a folder called `Cognome_Nome_IDpersona` with the following files:

- the two files `Edificio1.csv` and `Edificio2.csv`;
- one or more Python files that use the `mip` module to model and solve the problem; multiple scripts may be included, but the Python script to be executed *must* be named `main.py`.

The script `main.py` *must* accept one input argument, which must be a file in the same format as `Edificio1.csv` and `Edificio2.csv`.

The project evaluator *must* be able to execute your code by running these two terminal commands:

```
unzip Cognome_Nome_IDpersona.zip
python Cognome_Nome_IDpersona/main.py Cognome_Nome_IDpersona/Edificio1.csv
```

After solving the problem, the program *must* produce an output in which, for each trip i , the string `Viaggio i` is printed, followed by the sequence of points visited during trip i , from the base point (assumed to have index 0) to its return to the base, as in the example:

```
Viaggio 1: 0-4-11-17-...-2-0
Viaggio 2: 0-5-6-3-...-7-0
Viaggio 3: 0-9-...-0
Viaggio 4: 0-12-...-0
```

In addition, your script must produce a graphical representation of the drone's routes. Any error encountered by the evaluator in performing the steps above will result in penalties to your score.